

Report

Master's Programme in Data Science

Data Science Mini-Project Technical Report

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Tiivistelmä — Sammanfattning — Abstract

This report contains the technical information for the course project in the Data Science course. The first chapter presents the general idea of our project, the second chapter explains data collection and wrangling, the third presents our model and the fourth chapter concludes with reflections and lessons learned during the project.

ACM Computing Classification System (CCS)
General and reference

Avainsanat — Nyckelord — Keywords

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Chapter 1

Introduction

Our research is focused on discovering the extent of political polarization in the Finnish parliament between the years 2015-2026. We measure polarization by measuring how well we can identify individual speakers' party affiliation and how well these can be classified in distinct party groups based on the words and phrases they use. We also measure how much an individual MP differs in speech from their party average.

We leverage data from the Finnish parliament's open access database containing logs of every speech made for each parliamentary session in each term between 2015 and 2026. We further increase the robustness of our results by using the answers representatives used in their election compass prior to election, to see if there is a stronger party-line discipline in speeches made in parliamentary sessions compared to individual viewpoints expressed in the election compass.

Our goal is to create a method to measure political polarization, which can be re-used for each term or time period of interest by politically interested citizens and political journalists, to measure if Finnish parties are drifting closer or further from each other.

In chapter 2 of this tech report we describe the data we used for our research. We describe the data sources and explain how data was cleaned and formatted for further analysis. In chapter 3, we summarize the methods used to derive our results. Chapter 4 includes our reflections on the whole project, such as lessons learned, comparisons between the canvas and actual final deliverables as well as a report on our use of AI. The full code for the project is available in the team GitHub repository.

Chapter 2

Data collection and cleanup

The data used in this project was collected from the Finnish parliament's open access database. The data contains speeches made by members of parliament between the years 2015 and 2026, for a total of 133648 speeches. The data set contains identification fields, records on which matter the speech was made, the MP's party and position if in the current government as well as the raw text of the speech including possible interjections by other MPs ('välihuuto').

We add metadata on parliamentary term, party, governmental position, minister status (boolean) and speech type to the dataset. A parliamentary term is defined as the date when the new parliament convenes for the first time, party describes the speakers party at the time of the speech, governmental position defines if the speaker is a part of a current government party or in the opposition and minister status indicates if the speaker is currently serving as a minister.

The data is cleaned and stemmed following the methodology applied by Simola et al., 2023, in their research on previous Finnish parliamentary speeches. We chose this approach to maintain some comparability between previous research. However, we differ from previous research by not controlling for regional dialects or representative gender in our study, so results are not directly comparable but offer insights of similar style. Our approach is described in detail below.

We differ from Simola et al. (2023) by omitting any speeches made in Swedish as the total amount of speeches in Swedish is relatively low (1552) as well as those made by the representative of Aland Islands (270), which are mostly in Swedish as well. We further restrict the dataset to not include speeches made by parties of less than 5 representatives

per parliamentary term and individual MP's that do not belong to any party at the time of the speech.

We remove the interjections from the speeches, remove all references to party names or party abbreviations (SDP, KOK, PS etc.), remove all formalities, such as "Mr. Speaker...". Further, we remove all references to names as well. Names that have other meanings in Finnish are removed only from the middle of sentences, preserving the meaning of the words at the start of any sentence.

These decisions were made to ensure that the model we develop does not learn to recognize speakers based on names, choice of language or direct references to political parties. Formalities were omitted as these do not serve any useful purpose in classification as all parties use similar expressions, with the exception of the Swedish People's Party (RKP), which tends to use Swedish formalities. Small parties and independent MP's are excluded as they add noise to the model without offering meaningful data for the final analysis.

We also convert relevant symbols, €, % and §, to words as they can be important for topic identification. We remove stopwords from the speeches using the stop word list by Simola et al., 2023. We stem words using the Snowball stemmer from PyStemmer following the Simola et al. methodology. Removed names, parties or formalities are replaced by a stop-character "|", over which no bigrams can be formed, bigram formation is described more thoroughly in chapter 3.

We conduct an exploratory data analysis of the cleaned data, we find a total 130965 speeches after the restrictions described above, i.e. the restrictions cut 2683 speeches. The analysis confirms that in small parties, most speeches are held by a few MPs. The total amount of speeches made by party per term can be seen in table 2.1.

Table 2.1: Speeches by party by term

	2015–2019	**2019–2023**	**2023–2027**
VAS	4015	2769	2565
SDP	10577	8130	9962
VIHR	4046	3452	2772
KESK	11150	7738	5099
RKP	1185	964	630
KD	2462	1940	717
KOK	7575	8571	7868
SIN	1684	0	0
PS	7294	9413	8387

In the EDA we also find that parties differ by what topics most often appear in their speeches. Most speeches reference legislative proposals by the majority parties, however the secondary topics differ clearly between parties, as presented in table 2.2.

Table 2.2: Party topics excluding legislative proposals

Party	**Topic**
PS	immigration & crime
VIHR	climate & nature
KESK	regions & agriculture
VAS	low-income earners & austerity
SDP	employment
KOK	security & Nato
RKP	education
KD	euthanasia & marriage

Speech length by speech type is presented in table 2.3. Most speeches are actual speeches made by MP's or replies to other speeches, a total of 86,8% of total speeches. After cleanup, speeches that contain less than 20 words are only 0,7% of the dataset.

Table 2.3: Word length of speeches by speech type

speech type	**count**	**mean**	**min**	**max**
introductory	3841	471	17	1850
undefined	4231	365	6	1831
fast-paced	5518	413	3	875
group	16840	458	7	1111
actual	54771	291	3	2781
reply	61220	122	4	779

Chapter 3

Methodology

The methodology used in our research large follows the methodology of originally developed by Gentzkow et al., 2019 and as applied by Simola et al., 2023 to research in the Finnish parliamentary system. Our analysis of party separability relies on research by Gronow and Malkamäki, 2024 on how to measure separation between party blocs.

We form a feature matrix of phrases to be used in the analysis from the speeches. Phrases include singular words and bigrams formed from the speeches. We restrict both words and bigrams in phrase formulation by requiring that it must appear in at least 100 speeches, it must be used at least 10 times each parliamentary year and must exist in at least 10 of the parliamentary years. These restrictions yield a total of 26203 phrases, 13464 words and 12739 bigrams. We recognize that this will limit phrases that appear later in the data, especially for the latest parliamentary term.

We apply a TF-IDF weighting and use a multinomial logistic regression to classify speeches by party. Using TF-IDF we can increase the model accuracy by weighting words specific to a speech compared to common words. We attempt to limit bias and error in our classifier by applying several methods. We separate speeches into five subsets when training our model, four of the sets are used for training while the fifth is used for model validation. We limit the amount of speeches by party used in training to 300, to offset the weight of larger parties compared to small parties, eliminating party size bias. We also compare the results of our classifier to a leave-out-measure from Gentzkow et al. (2019).

We use subsampling between training runs, we randomly select 80% of party representatives per party for each run, for a total of 8 runs. We also apply a permutation test to create a baseline comparison for our model, in the permutation test the party labels are

mixed between representatives, the accuracy of this model should approximate a guess and the differentiability between parties should be 0.

We measure polarization through several measures. We measure (pairwise) separability between parties, measured as the probability of confounding the parties. Polarization is measured as the mean of separability between parties, a total of 28 pairwise measures of separability. We estimate a representative MP profile using the probability of party affiliation over 20 randomly selected speeches, measured over 2 sets, for a total of 40 randomly selected speeches.

We find that we are able to classify speeches to party blocs and that there is a difference in how different parties speak. Our model beats the leave-out-estimator, confirming that the model is able to classify the speeches better than a random guess. Full results are available in the final blogpost, available here. Full code and documentation is available in the project GitHub repository.

Chapter 4

Reflections

Our original canvas was very ambitious and we had big plans for our project, luckily many of those goals were determined to be stretch goals from the start. We were able to fill all the core propositions of the original canvas and answer the question we set to ourselves from the start with a relatively robust data science framework and we feel reasonably confident about our results.

We used AI extensively in the coding of the project itself which split opinions within the group. Some of us argued that the use of AI would detract from the learning experience while on the other hand, AI-enabled coding tools are widely used in industry and practice, indicated by this very course's guest lecture by Relex personnel. In the end, we agreed to use AI-enabled coding as long as we thoroughly vet the final results and understand everything about the process ourselves. Details on this verification step can be read in the use of AI section below. We feel that the discussions that took place within our group are ones that are shared in the wider world of data science, and expert professions overall, and that it was a valuable learning experience for us. We can clearly see the benefits of AI-assistance but still feel that it is critically important that we understand what we are doing instead of trusting an AI to do things for us. To not use the tools available to us would be inefficient and delay solving, sometimes critical, problems, however, a faulty solution is no solution at all either.

We also had some issues on how to delegate tasks between group members and designing a functional project timeline among the group. We fell into a silo-style workflow which led that not all team members worked on all aspects of the project, especially regarding coding. However, in the end we were able to reconcile and all team members participated in creating the final result. On a positive note, all major decisions were still made in a group setting even though the implementation was somewhat siloed.

Altogether, the project provided valuable insights on working in small teams to solve large problems and a lesson on how to better leverage the strengths of each individual team member in the process. We have also had the opportunity to learn more thoroughly about NLP and the unique challenges it poses for ML approaches, especially the steps necessary in wrangling written text datasets.

Use of AI in the report

We have used AI (Claude by Anthropic) in scoping the study, producing the code and figures for the study and to produce documentation for the code notebooks. The project Github repository contains logs of AI use in different stages of coding the model.

Results and outputs were verified by the team in detail. The team conducted a thorough code review so all team members understood and agreed on the code and verified the results. The team reviewed the code independently at first, listing questions in our shared group chat and then the team went through the results together during a team meeting before approving the final product together.

Microsoft Copilot was used for creating visuals in the group presentation.

Claude was used in formatting the visuals and to draft text for the final blog post, full commit logs are available on the project Github.

Claude was used to generate latex code for tables in the technical report, all text is human-made.

References

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